Encoding and Modulation

Topics Covered

1. Communication System Overview
2. Modulation – Definition & Types
3. Amplitude Modulation (AM)
4. Frequency Modulation (FM)
5. Phase Modulation (PM)
6. Digital Signals & Bit Rate
7. Digital-to-Digital Conversion
8. Line Coding Schemes
9. Block Coding (4B/5B, 8B/10B)

1. Communication System

Definition

A communication system involves the transfer of information or intelligence from a source to a receiver via a channel or medium. It consists of five fundamental blocks: the information source, transmitter, communication channel, receiver, and the end user.

Block Diagram Components

Source: Produces the information signal — can be analog or digital. Signals may be 1D (speech, music), 2D (images), 3D (video), or 4D (volume data over time).

Transmitter: Collection of circuits (transducer, amplifier, modulator, oscillator, mixer, antenna) that converts the message into a form suitable for channel propagation. It also increases the power level to cover large distances.

Channel: The medium over which the signal travels. Can be wired (twisted pair, coaxial cable, optical fibre) or wireless (satellite, microwave, Bluetooth).

Receiver: Performs the reverse of the transmitter — amplifies, demodulates, and recreates the original signal using antenna, amplifier, demodulator, and transducer.

User: The destination — a person, speaker, monitor, or computer that consumes the received information.

Primary Resources & Noise

Every communication system depends on two primary resources: **transmitted power** (average power of the transmitted signal) and **channel bandwidth** (band of frequencies allocated for transmission). Channels are classified as *power-limited* or *band-limited* based on whichever resource is the bottleneck.

Noise is the unavoidable presence of unwanted signals that disturb transmission. Sources include internal sources (electronic devices) and external sources (man-made interference, natural noise).

2. Modulation – Definition & Types

Definition

Modulation is the process of changing the characteristics (amplitude, frequency, or phase) of a high-frequency **carrier signal** in accordance with the amplitude of a low-frequency **modulating (information) signal**. The result is a *modulated signal* (bandpass signal) that is transmitted over the channel. At the receiver, the reverse process — **demodulation** — recovers the original baseband signal.

Why Modulation is Needed

- ✓ Reduces the size of the antenna required for transmission.
- ✓ Enables long-distance communication.
- ✓ Prevents mutual interference between different stations.
- ✓ Avoids mixing of signals from multiple sources.
- ✓ Reduces the effect of noise, improving reception quality.

Classification of Modulation

Modulation is broadly divided into **Analog Modulation** and **Digital Modulation**:

Category	Type	Schemes
Analog (Continuous)	Amplitude Modulation	AM, DSB-SC, SSB, VSB
	Angle Modulation	FM (NBFM, WBFM), PM (NBPM, WBPM)
Analog (Pulse)	Pulse Analog Modulation	PAM, PWM, PPM
Digital (Pulse)	Pulse Digital Modulation	PCM, DPCM, DM, ADM, ADPCM
Digital (Shift Keying)	Shift Keying Techniques	ASK, FSK, PSK, QAM

3. Amplitude Modulation (AM)

Definition

Amplitude Modulation (AM) is an analog-to-analog modulation technique in which the **amplitude of the carrier signal** is varied in proportion to the instantaneous amplitude of the modulating (message) signal, while the frequency and phase of the carrier remain constant.

Mathematical Representation

Given:

Modulating signal: $V_m(t) = V_m \sin(\omega_m t)$

Carrier signal: $V_c(t) = V_c \sin(\omega_c t)$

AM Signal: $V_{AM} = V_c [1 + m_a \sin(\omega_m t)] \sin(\omega_c t)$

Expanding:

$V_{AM} = V_c \sin(\omega_c t) + (m_a V_c / 2) \cos((\omega_c - \omega_m)t) - (m_a V_c / 2) \cos((\omega_c + \omega_m)t)$

Note: The three components are: Carrier Signal, Lower Sideband (LSB), Upper Sideband (USB).

AM Envelope

The **AM envelope** is the outline (shape) of the modulated signal. It contains all the frequencies that make up the AM signal. The envelope equation is:

$$s(t) = Ac\,[1 + k \cdot m(t)]$$

AM Spectrum & Bandwidth

The AM signal occupies a spectrum from (fc - fm) to (fc + fm). The two sidebands are:

- **Lower Sideband (LSB):** Band of frequencies between fc - fm(max) and fc.
- **Upper Sideband (USB):** Band of frequencies between fc and fc + fm(max).

$$\text{Bandwidth } (B) = [fc + fm(max)] - [fc - fm(max)] = 2\,fm(max)$$

Modulation Index (m)

The modulation index (also called depth of modulation or coefficient of modulation) quantifies how much the amplitude of the carrier is varied by the modulating signal.

$$m = Em / Ec$$

$$Em = \tfrac{1}{2}\,[Vmax - Vmin]\quad Ec = \tfrac{1}{2}\,[Vmax + Vmin]$$

$$\text{Percentage Modulation } M = (Em / Ec) \times 100$$

Condition	Value	Type
m = 1	100%	Critical / Full modulation
m < 1	e.g. 50%	Under modulation (preferred)
m > 1	> 100%	Over modulation (distortion occurs)

Advantages

- ✓ Simple circuit design and easy to implement.
- ✓ Compatible with simple envelope detector receivers.
- ✓ Widely used in AM radio broadcasting.

Disadvantages

- ✗ Carrier power is wasted (not information-bearing).
- ✗ Susceptible to amplitude noise and interference.
- ✗ Less power-efficient compared to DSB-SC or SSB.
- ✗ Over-modulation causes signal distortion.

4. Frequency Modulation (FM)

Definition

Frequency Modulation (FM) is an analog modulation technique in which the **frequency of the carrier signal** is varied in proportion to the instantaneous amplitude of the modulating signal. The peak amplitude and phase of the carrier remain constant. As the amplitude of the information signal changes, the carrier frequency changes correspondingly.

Key Characteristics

- The instantaneous change in carrier frequency is proportional to the amplitude of the modulating signal.

- A Voltage-Controlled Oscillator (VCO) is typically used to implement FM.
- FM is classified as Narrowband FM (NBFM) or Wideband FM (WBFM) based on modulation index.

FM Bandwidth (Carson's Rule)

$$B_{FM} = 2(1 + \beta)B$$

where β is the modulation index (ratio of frequency deviation to modulating signal frequency) and **B** is the bandwidth of the audio signal.

Advantages

- ✓ Better noise immunity than AM.
- ✓ Higher fidelity audio quality.
- ✓ Less susceptible to amplitude-based interference.

Disadvantages

- ✗ Requires larger bandwidth than AM.
- ✗ More complex transmitter and receiver circuits.
- ✗ Signal quality degrades sharply below a threshold (capture effect).

5. Phase Modulation (PM)

Definition

Phase Modulation (PM) is an analog modulation technique in which the **phase of the carrier signal** is varied in proportion to the instantaneous amplitude of the modulating signal. The amplitude and frequency of the carrier remain constant. PM and FM are closely related; in FM the frequency change is proportional to signal amplitude, while in PM the frequency change is proportional to the *derivative* of signal amplitude.

PM Bandwidth

$$B_{PM} = 2(1 + \beta)B$$

The bandwidth formula is the same form as FM, but here β depends on both the bandwidth and maximum amplitude of the modulating signal.

FM vs PM Comparison

Parameter	FM	PM
What changes	Carrier frequency	Carrier phase
Proportional to	Amplitude of m(t)	Derivative of m(t)
Constant parameters	Amplitude, Phase	Amplitude, Frequency
Bandwidth formula	2(1+β)B	2(1+β)B
Circuit element	VCO directly	Differentiator + VCO

6. Digital Signals & Data Rate Concepts

Digital Signals

A digital signal uses discrete voltage levels to represent binary data (0s and 1s). The number of bits per level determines how many voltage levels are needed. For L levels, each level represents $\log_2 L$ bits. Example: 2 levels → 1 bit/level; 4 levels → 2 bits/level.

Key Definitions

Bit Rate (N): Number of bits transmitted per second (bps). E.g., 100 pages/sec × 24 lines × 80 chars × 8 bits = 1.536 Mbps.

Bit Length: Physical distance one bit occupies on the medium. Bit length = Propagation speed × Bit duration.

Throughput: Actual data delivery rate of a network. Always ≤ bandwidth.

Latency (Delay): Total time for a message to travel from source to destination. Latency = Propagation time + Transmission time + Queuing time + Processing delay.

Propagation Time: Time for a bit to travel from source to destination. = Distance / Propagation Speed.

Transmission Time: Time to push all bits of a message onto the link. = Message size / Bandwidth.

Queuing Time: Variable time a message waits at intermediate nodes before processing. Increases with network load.

Bandwidth-Delay Product: Defines the number of bits that can fill the link at any given time. = Bandwidth × Delay.

Data Rate vs Signal Rate

The **data rate** (bit rate, N) is the number of data bits sent per second. The **signal rate** (baud rate, S) is the number of signal elements (symbols) per second. The relationship between them is:

S = N / r (r = data elements per signal element)

S_ave = c × N × (1/r) baud (c = case factor)

Goal: increase data rate while *decreasing* signal rate to save bandwidth.

7. Digital-to-Digital Conversion

Overview

Converting digital data to digital signals involves three techniques: **Line Coding** (always required), **Block Coding** (optional), and **Scrambling** (optional). Line coding is the fundamental process; block coding and scrambling add redundancy and synchronisation capabilities.

Line Coding — Definition

Line coding is the process of converting a sequence of binary data bits into a digital signal for transmission. At the sender, an encoder maps bits to voltage levels (signal elements); at the receiver, a decoder reverses the process to recover the original bits. The signal carries **data elements** (bits) using **signal elements** (the smallest unit of a digital signal, timewise).

8. Line Coding Schemes

8.1 Unipolar Scheme — NRZ

In a unipolar scheme, all signal levels are on *one side* of the time axis (either all positive or all at zero). The simplest form is **Non-Return-to-Zero (NRZ)**, where a positive voltage represents bit 1 and zero voltage represents bit 0. The signal does not return to zero at the middle of each bit. Normalized power = ½ V^2.

Advantages	Disadvantages
✓ Very simple to implement. ✓ Good signal rate efficiency.	✗ DC component causes baseline wander. ✗ Poor synchronisation with long runs of identical bits. ✗ Not suitable for channels with poor DC performance.

8.2 Polar Schemes — NRZ-L and NRZ-I

Polar schemes use voltages on **both sides** of the time axis (positive and negative). Two variants exist:

- **NRZ-L (NRZ-Level):** The voltage level directly determines the bit value. Positive = 1, Negative = 0 (or vice versa). S_ave = N/2.
- **NRZ-I (NRZ-Invert):** A transition (inversion) in voltage level at the start of a bit represents 1; no transition represents 0. S_ave = N/2.

Note: NRZ-I is better than NRZ-L because it is immune to polarity reversals in the cable.

Advantages	Disadvantages
✓ Simple implementation. ✓ Good signal rate (S = N/2). ✓ NRZ-I immune to polarity reversal.	✗ Synchronisation problem with long identical-bit sequences. ✗ DC component present in NRZ-L. ✗ No error detection capability.

8.3 Return-to-Zero (RZ)

RZ uses three voltage levels: positive, negative, and zero. The signal transitions to zero in the *middle* of each bit period, providing built-in synchronisation. Bit 1 starts positive and returns to zero; bit 0 starts negative and returns to zero. S_ave = N (twice the signal rate of NRZ).

Advantages	Disadvantages
✓ Built-in synchronisation — no long runs without transitions. ✓ DC-balanced with equal 1s and 0s.	✗ Requires two signal changes per bit → wastes bandwidth. ✗ Needs larger bandwidth than NRZ. ✗ More complex circuitry.

8.4 Biphase — Manchester & Differential Manchester

Biphase schemes combine the ideas of RZ (mid-bit transition) and NRZ.

- **Manchester Encoding:** The bit duration is divided into two halves. A high-to-low transition in the middle represents 1; a low-to-high transition represents 0 (or reversed, depending on convention). Combines RZ + NRZ-L concepts. S_ave = N.
- **Differential Manchester:** There is always a transition in the middle of the bit for synchronisation. The bit value is determined by the presence or absence of a transition at the beginning of the bit: transition = 0, no transition = 1. Combines RZ + NRZ-I concepts. S_ave = N.

Note: Both Manchester schemes have minimum bandwidth = 2 × NRZ bandwidth. They are used in Ethernet (10Base-T uses Manchester).

Advantages	Disadvantages
✓ Excellent synchronisation (guaranteed mid-bit transition). ✓ No DC component. ✓ Simple error detection possible. ✓ Differential Manchester immune to polarity reversal.	✗ Requires twice the bandwidth of NRZ. ✗ Higher signal rate (S_ave = N) uses more channel resources.

8.5 Bipolar Schemes — AMI & Pseudoternary

Bipolar encoding uses three voltage levels: positive, negative, and zero. The zero level encodes one bit value while the other bit value alternates between positive and negative.

- **AMI (Alternate Mark Inversion):** Bit 0 → zero voltage; Bit 1 → alternating positive/negative voltage. S_ave = N/2.
- **Pseudoternary:** The opposite of AMI. Bit 1 → zero voltage; Bit 0 → alternating positive/negative voltage. S_ave = N/2.

The alternating polarity of the non-zero pulses provides a **built-in error detection** mechanism — two consecutive same-polarity pulses (a bipolar violation) indicate an error.

Advantages	Disadvantages
✓ No DC component. ✓ Same signal rate as NRZ but better spectral properties. ✓ Built-in single-error detection via bipolar violation. ✓ Better suited for AC-coupled channels.	✗ Long strings of 0s (AMI) cause synchronisation loss. ✗ Receiver must distinguish three levels, requiring more complex circuitry.

8.6 Multilevel Schemes — mBnL

Multilevel schemes encode **m data bits** as **n signal elements** with **L voltage levels**. The goal is to increase data rate or reduce required bandwidth. The constraint is: $2^m \leq L^n$. These are denoted **mBnL** (m Binary bits encoded as n L-level elements).

2B1Q (Two Binary, One Quaternary):

Two data bits are encoded as one quaternary (4-level) signal element. Levels: -3, -1, +1, +3. Mapping: 00→-3, 01→-1, 10→+3, 11→+1. S_ave = N/4 — achieves 4× bandwidth efficiency over NRZ-L. Used in DSL (Digital Subscriber Line) technology.

8B6T (Eight Binary, Six Ternary):

8 data bits → 6 ternary (-/0/+) signal elements. Used with 100BASE-4T cable. 2^8=256 patterns mapped to 3^6=729 signal patterns, leaving 473 redundant patterns for synchronisation and error detection. DC balance maintained via disparity control.

4D-PAM5 (Four-Dimensional, Five-Level PAM):

Data is sent simultaneously over **4 wires** using 5 voltage levels (-2, -1, 0, +1, +2). Level 0 is reserved for forward error detection. Achieves 1 Gbps by sending 250 Mbps on each of 4 wires at 125 MBd. Used in Gigabit Ethernet (1000BASE-T).

Advantages

- ✓ Higher data rate for the same bandwidth.
- ✓ Better bandwidth efficiency than simple schemes.
- ✓ Can incorporate error detection (redundant patterns).

Disadvantages

- ✗ More complex circuits needed to distinguish multiple signal levels.
- ✗ Less noise immunity — smaller gap between levels.
- ✗ Reduced redundancy in some schemes (2B1Q has none).

8.7 Multitransition — MLT-3

MLT-3 uses three voltage levels (+V, 0, -V) and cycles through them sequentially: +V → 0 → -V → 0 → +V. A transition occurs for bit 1; no transition for bit 0. The signal rate is one-third of NRZ. Used in 100BASE-TX Fast Ethernet (FDDI).

9. Block Coding

Definition

Block coding changes a block of **m bits** into a block of **n bits** (n > m), adding redundancy for synchronisation and error detection. It is denoted **mB/nB** (e.g., 4B/5B). The process involves three steps: (1) *Division* — the bit stream is split into m-bit groups; (2) *Substitution* — each m-bit group is replaced by a unique n-bit code from a lookup table; (3) *Combination* — the n-bit groups are combined into a new stream with more bits than the original.

4B/5B Block Coding

The **4B/5B** scheme encodes every 4-bit group as a unique 5-bit code. It is designed to be used with NRZ-I line coding. NRZ-I alone has a synchronisation problem with long sequences of 0s. 4B/5B ensures no encoded 5-bit group has more than one leading zero or more than two trailing zeros — so the combined stream never has more than **three consecutive 0s**.

- 2^4 = 16 data patterns mapped to 16 of the 2^5 = 32 possible 5-bit codes.
- The 16 unused codes are used for control sequences (Idle, Halt, Start/End delimiters, etc.) or declared invalid.
- The actual data throughput is 80% (4 data bits per 5 transmitted bits).

Note: 4B/5B + NRZ-I is used in 100BASE-TX (Fast Ethernet) and FDDI.

Advantages

- ✓ Eliminates long runs of 0s — solves NRZ-I synchronisation problem.
- ✓ Redundant codes provide error detection.
- ✓ Relatively simple lookup tables.

Disadvantages

- ✗ 20% bandwidth overhead (5 bits sent for every 4 data bits).
- ✗ No DC balance (still relies on NRZ-I for this).

8B/10B Block Coding

The **8B/10B** scheme encodes every 8-bit group as a 10-bit code. It is implemented as a combination of **5B/6B** encoding (for the 5 most significant bits) and **3B/4B** encoding (for the 3 least significant bits). A **disparity controller** tracks the running difference between 0s and 1s — if the current block would increase existing disparity, the entire code is complemented (all bits flipped). This achieves DC balance.

- $2^{10} - 2^{8} = 768$ redundant codes available for error detection.
- Actual throughput: 80% (8 data bits per 10 transmitted bits).
- Used in Gigabit Ethernet, Fibre Channel, USB 3.0.

Advantages	Disadvantages
✓ Excellent built-in error detection (768 redundant patterns).	✗ 20% bandwidth overhead.
✓ Guaranteed DC balance via disparity controller.	✗ More complex implementation than 4B/5B.
✓ Strong synchronisation properties.	✗ Requires tracking running disparity.
✓ Superior to 4B/5B overall.	

Quick Reference — Line Coding Comparison

Scheme	Levels	Signal Rate	DC Component	Synchronisation
Unipolar NRZ	2	N/2	Yes	Poor
Polar NRZ-L	2	N/2	Possible	Poor
Polar NRZ-I	2	N/2	Possible	Poor
Polar RZ	3	N	No	Good
Manchester	2	N	No	Excellent
Diff. Manchester	2	N	No	Excellent
Bipolar AMI	3	N/2	No	Poor (0s)
2B1Q	4	N/4	Possible	Moderate
MLT-3	3	N/3	No	Moderate

Quick Reference — Analog Modulation Comparison

Type	What Changes	Constants	Bandwidth	Noise Immunity
AM	Amplitude	Frequency, Phase	2 fm(max)	Low
FM	Frequency	Amplitude, Phase	2(1+β)B	High
PM	Phase	Amplitude, Frequency	2(1+β)B	High

These notes are compiled from Lecture 05-07: Encoding and Modulation. All topics from the lecture have been covered, including Communication Systems, AM/FM/PM Modulation, Digital Signals, and all Line Coding and Block Coding schemes.